NETRAJYOTI AI

Reliable Bengali eye-care guidance

Clear Bengali guidance
for your next eye-care step

Bengali-first interaction
for rural West Bengal

How can reliable health information reach people who do not use English or have limited digital access?

NetraJyoti helps people identify a safe next step without diagnosing or replacing clinical care.

Fixed Bengali selections → **Deterministic route** → **Clear next action**

The access problem

Language and service uncertainty can delay a safe eye-care decision

THE DECISION GAP

People need answers before they can act

A Bengali-speaking user may struggle to decide whether care is urgent, which guidance is trustworthy, or where to seek help.

1. **Is this urgent?**
   Symptoms may feel unclear
2. **Can I trust this?**
   Information may not be Bengali
3. **Where should I go?**
   Local service direction may be unclear

Design response: provide a Bengali-first guided selection journey, then present a safe next step through a controlled route.

Route selection and Bengali explanation

The Java deterministic engine selects the care route with PostgreSQL. Ollama explains only after the route lock.

ROUTE SELECTION

Bengali UI
Fixed symptom selections

→

Java deterministic engine
Validates submitted codes

→

PostgreSQL
Matches approved routing criteria

→

Route lock
Urgent, routine, or human support

ROUTE LOCK | Guidance and model output cannot change the selected care route

Optional voice and free-text notes provide context only. They do not determine the route.

POST-ROUTE EXPLANATION

PostgreSQL RAG
Approved guidance for the locked route

→

Local Ollama
If enabled and available, creates a Bengali explanation

→

Java deterministic engine
Safe explanation or approved fallback

Ollama receives approved guidance and never decides, changes, or diagnoses the route.

User journey

A single Bengali-first flow moves the user from consent to a clear next action

1 → 2 → 3 → 4 → 5

Consent

Clear purpose and data choice

Symptoms

Fixed Bengali options or voice input

Safety check

Urgent signs checked first

Safe route

Urgent, routine, or human support

Next action

Guidance, service direction, sharing

Every result offers a Back or Start again option so people can revisit choices without losing the journey.

Application technology stack

A lightweight web client, Java deterministic engine, PostgreSQL data layer, and local Ollama model

Experience layer

React PWA

TypeScript, Vite, Tailwind CSS

API and route engine

Java deterministic engine

Java 21, Spring Boot 3, REST API

Clinical data and RAG

PostgreSQL

Approved criteria, guidance, pathways, and source links

Language explanation

Local Ollama

Default model: llama3.2:3b
Bengali explanation or approved fallback

Docker Desktop runs the PostgreSQL and Ollama services for the local demonstration environment.

Current safety boundary

The live MVP separates route selection from optional generated explanation.

DETERMINISTIC ROUTE

Java deterministic engine selects the care route

It uses fixed UI codes and approved PostgreSQL routing criteria to determine urgent, routine, or human support.

Route lock: no later stage can override it

CONDITIONAL EXPLANATION

Ollama explains the locked route

When the AI-summary switch is on, Local Ollama is reachable, and approved guidance plus a safe response are available, it creates Bengali explanation. Otherwise the Java deterministic engine displays an approved fallback.

Explanation quality cannot affect the route

Current MVP: no model-led diagnosis, treatment recommendations, or route selection.

Future governed AI-agent scope

Future agents support administration under human review. They do not control safety routing.

Boundary: agents will not autonomously route users, diagnose conditions, prescribe treatment, or publish unverified information.

Service directory

Agent checks proposed updates for completeness

Change owner: Vision centre or eye-clinic staff

Publisher: Product and operations administrator

Bengali content

Agent prepares drafts from approved sources

Approver: Clinical reviewer or content administrator

Content records

Agent assists with ownership and version records

Owner: Product and operations administrator

Terms: service directory = local facility details, not patient records. Bengali content = reviewer-approved guidance. Content records = governance metadata: owner, source, version, approval, and change history.

Future IVR: confirmed menu selections map to fixed codes.
Future free text and medical documents: post-route context only. They cannot change the route.

Personas and pilot usage measures

Pilot measurement should show who uses the service, where demand occurs, and how the journey performs.

APPLICATION PERSONAS

- **Community member**
  Bengali-language user seeking a safe next step
- **Support person**
  ASHA worker or family helper supporting the journey
- **Service staff**
  Vision centre or eye-clinic staff maintaining facility details
- **Clinical reviewer**
  Approves criteria and Bengali guidance
- **Operations administrator**
  Controls versions, publication, and releases

PILOT USAGE MEASURES

- **Location**
  Sessions by district, block, and town
- **Demand**
  Time-of-day and day-of-week traffic
- **Route mix**
  Urgent, routine, and human-support outcomes
- **Access mode**
  Voice versus text completion rate
- **Experience**
  Result latency and approved fallback rate

These are proposed pilot measures. The current MVP has no production usage baseline.

Production data lifecycle

Approved clinical sources become versioned routing criteria and guidance before the application can use them.

DATA COMPONENT	RUNTIME USE	CLINICAL REVIEW AND SOURCE MATERIAL
Pathways and routing criteria	Java deterministic engine selects the route	Clinical reviewer signs off precedence and scenario tests. **SOURCE MATERIAL** **WHO VESIH · WHO PECI · WHO Eye Care in Health Systems · India DGHS / NPCBVI**
Guidance and source links	PostgreSQL RAG retrieves approved guidance	Clinical reviewer approves Bengali guidance and sources; version is retained before publication.
Service directory	Result UI shows local service direction	Vision centre or eye-clinic staff propose facility facts; operations administrator publishes verified changes.
Content records	Administration, audit, rollback, and traceability	Governance metadata only: owner, source, version, approval status, and change history. Not patient clinical records.

Lifecycle: authoritative source → clinical draft → clinical review and sign-off → versioned PostgreSQL publication → route-validation tests → monitored release or rollback.

Current MVP uses static demo content. A production release requires a governed approval, validation, release, and rollback process.

Population reach and pilot scope

Latest detailed Census baseline for potential reach. Pilot scope requires explicit design decisions.

TAM

97.2M

Bengali mother-tongue speakers in India

Census 2011 detailed baseline

SAM

18.2M

Residents in North and South 24 Parganas

Census 2011 district baseline

SOM

Not yet set

Requires pilot sites and eligible participants

No forecast before pilot design

Census 2027 is scheduled. Use refreshed local population data when the pilot scope is approved.

Official sources: Office of the Registrar General & Census Commissioner, India (ORGI), Census 2011 C-16 Population by Mother Tongue (catalogue 10191); District Census Handbooks: North 24 Parganas (1349) and South 24 Parganas (1362); Census 2027 Gazette notification, censusindia.gov.in.

What the MVP establishes

A practical foundation for safer access to Bengali eye-care guidance

Bengali-language access

Voice and text options
with visible fallback

Fixed safety route

Java deterministic engine
applies approved criteria

Clear next action

Guidance connects people
to service direction

The product guides users toward the next step while clinical judgment and content approval remain with people.

Foundation for supervised validation and controlled expansion

Live demonstration

A short walkthrough of the user journey and the locked safety-route decision

DEMO FLOW

1

Start the Bengali journey

Select Bengali options

2

Select symptoms

Fixed selections determine the route

3

View the safe result

Show route, guidance, and next action

Demo check: the Java deterministic engine fixes the route before PostgreSQL RAG and Local Ollama provide the explanation.

Thank you

Questions and discussion

Reliable Bengali eye-care guidance with a fixed safety route

Fixed Bengali selections · approved clinical content · clear next action